Latest Wireless Technologies, Devices And Protocols For SECURE HOME AUTOMATION

DEEPSHIKHA SHUKLA

A complete wireless networking is an indispensable element of home automation. In this article we discuss the latest wireless technologies that are making our homes safer and more secure

Wireless technology for smart homes helps you keep an eye on your home when you are away. You can protect what is valuable to you at home or even manage your business remotely. It provides a safe, convenient and smart life through intelligent devices, cloud-based platforms and advanced technologies.

Innovative wireless technologies including 4G and 5G have paved the way for home automation. Now, you can enjoy home entertainment and remotely protect and take care of your family in a smarter and easier way. The Internet of Things (IoT) enables you to control your appliances, sharing videos or images captured by cameras to friends and families, turning on alarms and sensors, from anywhere. It allows your to do so by way of controlling programmable timing controllers, electrical appliances control systems and more via apps on your smartphone.

Latest technologies for better connectivity

LTE Small Cell technology provides complete, strong in-home wireless network coverage to support the implementation of home automation. It provides cellular indoor coverage via broadband IP backhaul. The technology can solve cellular network issues including weak coverage, low data rate and site acquisition. The solution increases the capacity for data service and offloads macro network by absorbing traffic from indoor hotspots. It gives quality connectivity and, hence, an excellent home automation experience for households.

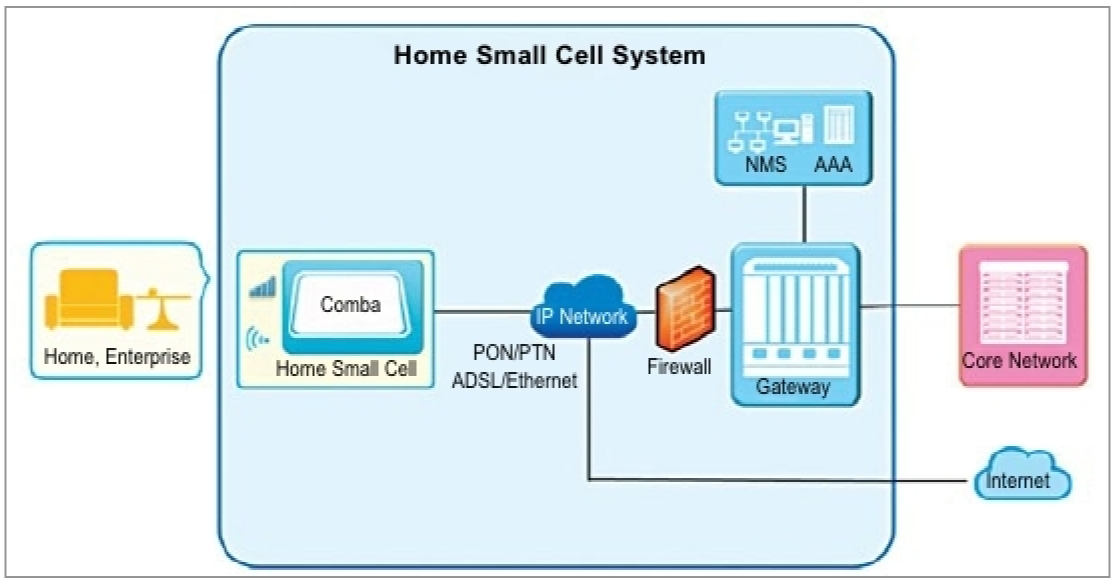

Fig. 1: Topology of home cell solution (Credit: Comba Telecom)

Marie Ma, senior director of technical marketing solutions and general manager of enterprise business, Comba Telecom, says, "Comba LTE Small Cell solution includes home small cells, gateways and a network management system (NMS). It extends LTE mobile coverage to small, hard-to-reach areas. The solution supports up to eight active users and 24 RRC connected users. It is ideal for small homes and offices, and small public hotspots." The architecture of LTE Small Cell solution is shown in Fig. 1.

Dhananjay Sharma, chief operating officer, SenRa Tech Pvt Ltd, says, "LoRaWAN as a wireless technology is extensively used in home automation. The application ranges from door/window motion sensing, temperature sensing, carbon monoxide monitoring, water metering, gas metering and more. LoRaWAN gateways with integrated Wi-Fi are already available. This will further augment the use cases for high data rates as well as low data devices and battery-powered sensors."

Bluetooth Low Energy (BLE) is globally interoperable on 2.4GHz band, and is available on all smartphones and tablets. Recent standard evolutions with Bluetooth 5 as well as BLE Mesh make the technology even more suitable in a smart home. Bluetooth 5 gives double data throughput (2Mbps), which means that communication through an app is faster.

BLE also results in more power-efficient solutions. For example, a battery-driven sensor can send data much faster and then go to sleep, thus saving battery. Bluetooth 5 also increases coverage as the range between devices can be four times that of BLE 4.2, all due to the new BLE long-range coding scheme.